Bayesian Analysis - Final Assignment
Bayesian modeling and comparison of running performance

Ares Inglada Grañó and Liliu Martínez Xicoira

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1 Abstract

This project analyzes marathon finishing times in the Boston Marathon using Bayesian hierarchical models. A dataset containing 40,902 observations from 15,803 runners between 2014 and 2019 was compiled and preprocessed, including the identification of repeated participants across race editions. Exploratory analyses revealed important effects of age, gender, country, and race year on performance. Three Bayesian models of increasing complexity were developed, incorporating year, country, and runner-level random effects. Results show that hierarchical structures substantially improve the explanation of performance variability, with runner-specific effects providing the largest contribution. The final model achieved strong predictive performance and highlights the value of Bayesian hierarchical modeling for analyzing longitudinal sports data.

2 Introduction

2.1 Description of the Problem and Justification for its Selection

Running is one of the most widely practiced endurance sports worldwide. Understanding the factors associated with marathon performance is relevant for athletes, coaches, and researchers interested in human performance and sports analytics.

The Boston Marathon is one of the oldest and most prestigious marathon events, providing a large amount of publicly available historical data. Its long tradition and international participation make it an ideal case study for investigating marathon finishing times.

Bayesian statistical methods are particularly suitable for this problem because they allow uncertainty quantification and provide a flexible framework for modeling variability across individuals and race editions.

2.2 Objectives and Scope

The objective of this project is to analyze marathon finishing times using Bayesian hierarchical models.

The specific goals are:

- To study the relationship between runner characteristics and finishing time.
- To quantify variability between race editions, countries, and individual runners.
- To develop and compare Bayesian models for explaining and predicting marathon performance.
- To evaluate the uncertainty associated with model estimates and predictions.

The analysis is based on Boston Marathon results from 2014 to 2019. After preprocessing, the final dataset contains 40,902 observations from 15,803 runners representing 61 countries.

A key characteristic of the dataset is the presence of repeated observations for many runners, motivating the use of hierarchical models with runner-level effects. The analysis focuses on variables consistently available across all years: age, gender, country, race year, runner identifier, and official finishing time.

2.3 Data Sources and Preprocessing

The Boston Marathon is a well-established race with a well-documented history of its participants. The official Boston Athletic Association (BAA) website [1] provides detailed information about runners and race results for each edition of the event: name, city, country, age, gender, division, time for each 5k split, finishing time and overall and division rank.

Although the BAA website contains rich information, its structure makes automated data collection difficult. Therefore, this study relies on publicly available datasets obtained from Kaggle [6] and GitHub repositories [7], which were originally compiled from official BAA data.

Depending on the marathon year's data results, several variables contained a large proportion of missing values. For example, while some editions included split times at intermediate checkpoints, other years had these fields almost entirely missing. As a result, split times were excluded from the analysis to ensure consistency across years.

After standardizing variable names and formats, all yearly datasets were concatenated into a single database. During this process, it became apparent that many runners participated in multiple editions of the Boston Marathon. To account for repeated observations, a unique runner identifier (ID) was created.

The runner identity was reconstructed using the combination of name, gender, city, and birth year. Birth year was estimated from the runner's age and the race year. This procedure allowed the identification of repeated participants across different marathon editions.

Table 1 presents an example of the final dataset used in this study.

Table 1: Sample of observations from the final dataset (2014–2019)

Name	Age	M/F	City	Country	Pace	Official Time	overall_result	gender_result	division_result	Year	Birth Year	ID
MARCHIO, JENNIFER A.	35	F	BRIGHTON	USA	0:10:23	4:31:58	23121	10054	5000	2015	1980	22912
REHRIG, JIM	71	M	NAZARETH	USA	0:10:23	4:32:00	23123	13068	59	2015	1944	22914
CLARK, THANH	50	F	CARMEL	USA	0:10:23	4:32:02	23125	10056	1036	2015	1965	22916
MATROS, MICHAEL A.	30	M	MARLBORO	USA	0:10:23	4:32:04	23127	13071	4535	2015	1985	22918
MARCHIO, JENNIFER A.	34	F	BRIGHTON	USA	0:10:50	4:44:22	25402	10956	1445	2014	1980	22912

2.4 Response Variable and Explanatory Variables

The response variable considered in this study is the official marathon finishing time. Since marathon times are strictly positive and exhibit a right-skewed distribution, the logarithm of the finishing time was used as the response variable in all Bayesian models.

Several explanatory variables were selected based on their availability across all years and their relevance for marathon performance. These include runner age, gender, country of origin and race year. In addition, a unique runner identifier was created to account for repeated participations across multiple marathon editions.

Table 2 summarizes the variables used in the analysis.

Table 2: Variables used in the analysis

Variable	Description
Age	Standardized runner age
M/F	Runner gender (Male/Female)
Country	Runner country
Year	Marathon year
ID	Runner identifier
log_time	Log-transformed official finishing time

The variable `log_time` is treated as the response variable in the Bayesian models. Age, gender, year, country, and runner identifier are considered explanatory variables or grouping factors, depending on the model specification.

3 Data and Exploratory Analysis

3.1 Exploratory Data Analysis

Before fitting the Bayesian models, an exploratory analysis was conducted to understand the main characteristics of the dataset and the relationships between the variables of interest.

The dataset consists of Boston Marathon results from 2014 to 2019 and exhibits a natural hierarchical structure, with observations nested within runners, countries, and race years. Since many runners participated in multiple editions of the race, the dataset exhibits a natural hierarchical structure with observations nested within runners, countries, and race years.

The exploratory analysis focused on:

- The distribution of marathon finishing times.
- Differences in performance across age groups and gender.
- Variability between race years.
- Variability between countries.
- The identification of repeated runners participating in multiple editions.

The results of this analysis provided empirical support for the use of hierarchical Bayesian models, motivating the inclusion of year-level, country-level and runner-level random effects in the subsequent modeling stages.

3.1.1 Exploratory finishing times analysis

First, the distribution of finishing times was studied in order to better understand its behavior. Figure 1 exhibits "heavy tails," with a small number of elite athletes performing significantly better than the mean, and a wide spread among recreational runners. So as to correct this, a logarithmic transformation was applied to finishing times, resulting in Figure 2. This created a more symmetric distribution and stabilized the variance, allowing for a fair comparison across all performance levels.

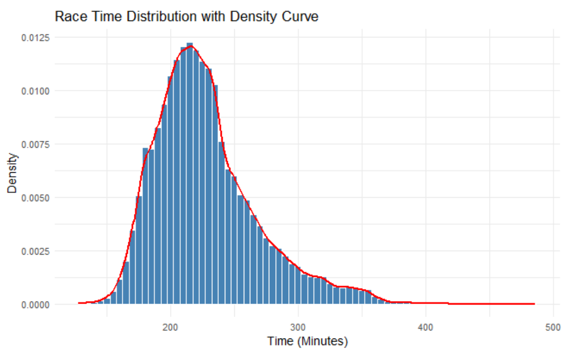

Figure 1: Distribution plot of finishing times (minutes)

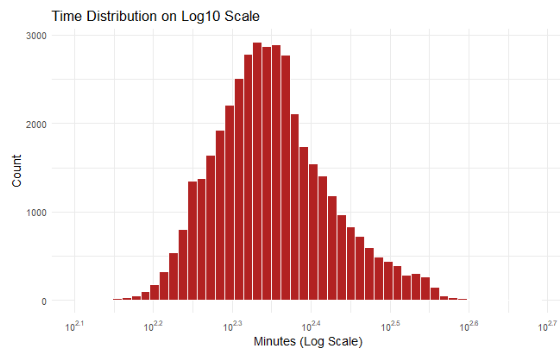

Figure 2: Distribution plot of transformed logarithmic finishing times (minutes)

3.1.2 Exploratory age analysis

The relationship between age and marathon performance is a central component of this study. As shown in Figure 3, although it ranges from 18 to 80, the demographic of the Boston Marathon participants is predominantly middle-aged.

To explore the impact of age on performance, Figure 4 illustrates the relationship between age and official finishing times. The scatter plot reveals a non-linear relationship, which is highlighted by the red smoothing line. This highlights there is a peak performance centered around the mid-30s. Nonetheless, starting approximately at age 40, there is a clear and consistent upward trend in finish times.This accelerates slightly in the later decades of life.

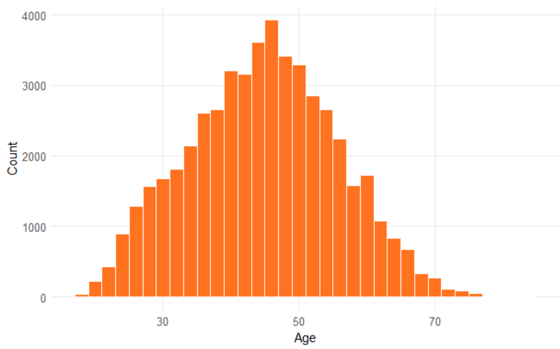

Figure 3: Histogram of overall marathon runner's age

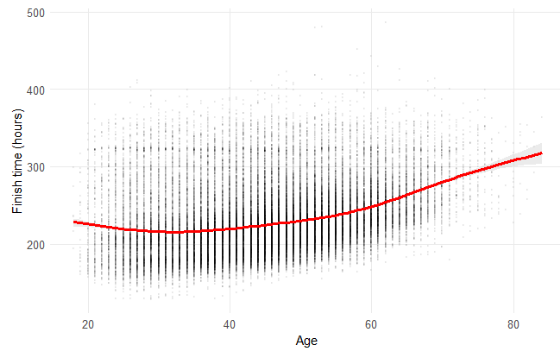

Figure 4: Scatterplot of finishing time by age

This exploratory analysis confirms that age is a crucial predictor of performance but suggests that a simple linear relationship may not fully capture the biological reality of aging and endurance.

3.1.3 Exploratory gender analysis

To examine the influence of biological sex on performance, Figure 5 presents a combined violin and box-plot of finishing times for male (M) and female (F) participants.

The distributions reveal a clear performance gap between the two groups, where the median finishing time for male runners is noticeably lower (faster) than that of female runners. Specifically, the male distribution is centered closer to the 210-minute mark, while the female distribution centers around 235-minutes.

Despite the difference in central tendency, both groups exhibit a similar unimodal shape with a significant right-skew. This "heavy tail" due to the large amount recreational runners. The violin plot also shows that the density of elite-level performances reaches lower absolute values for men than for women. This gender-based variation confirms that gender is a fundamental covariate that must be included in any predictive model for marathon performance.

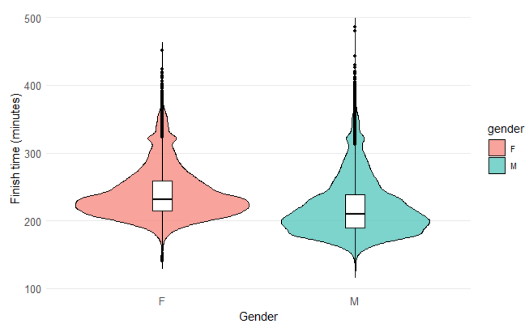

Figure 5: Distribution plot of finishing time by gender

3.1.4 Exploratory country analysis

The geographic diversity of the Boston Marathon is illustrated in Figure 6, which highlights extreme heterogeneity in performance across different nations. This Figure shows that countries such as Ethiopia and Kenya tend to exhibit lower median finishing times than the overall average. In contrast, countries with the highest number of participants, such as the USA, Canada (CAN), and Mexico (MEX), show much wider distributions with a vast number of outliers on the slower end of the spectrum.

This high degree of variability between countries indicates there are differences in performance levels across different nationalities and emphasizes the importance of including this grouping variable when studying finishing times.

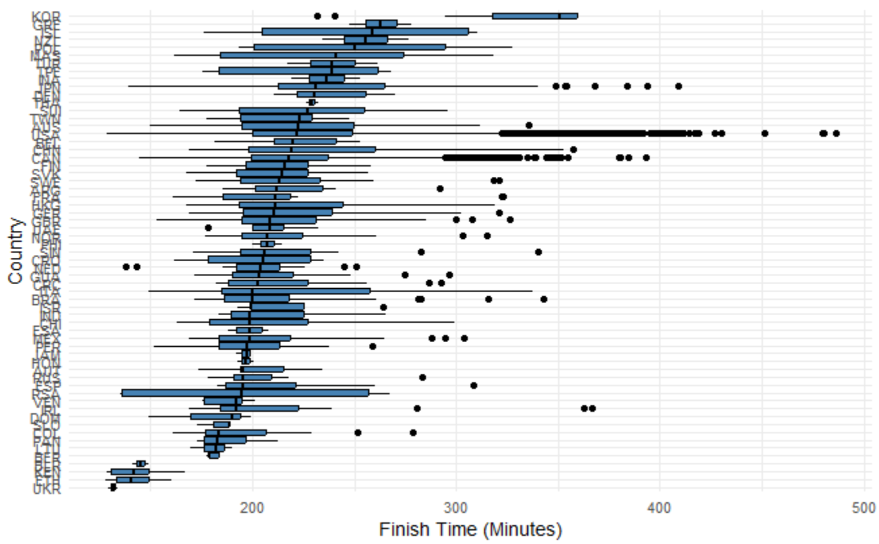

Figure 6: Box-plot of finishing time of each country

3.1.5 Exploratory Analysis of Repeated Runners

One of the key features of the dataset is the presence of runners who participated in multiple editions of the Boston Marathon. Since repeated observations from the same athlete are likely to be correlated, this longitudinal structure is important for the modeling process.

After reconstructing runner identities, 15,803 unique runners were identified, generating 40,902 observations between 2014 and 2019. Figure 7 shows the distribution of marathon participations per runner. Most runners participated in two editions (10,055 runners), and 382 runners appeared in all six editions included in the study.

This repeated-participation structure provides longitudinal information and motivates the inclusion of runner-level random effects. By accounting for individual-specific variability, Model 3 can distinguish persistent differences between runners from variation attributable to age, country, and race year.

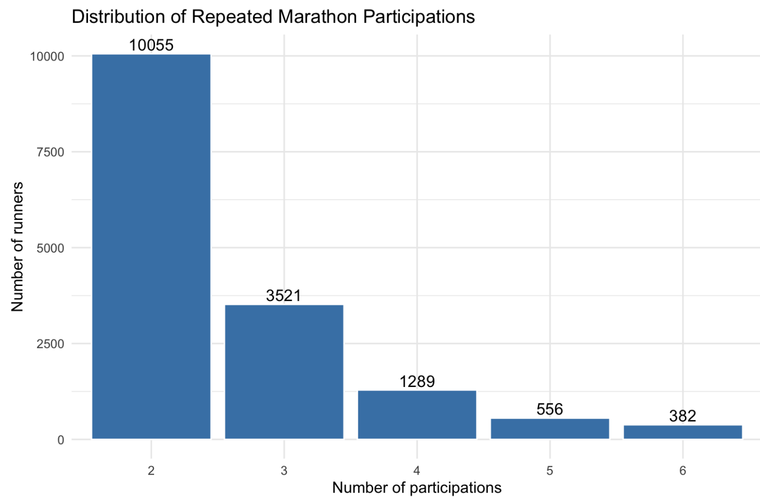

Figure 7: Distribution of the number of Boston Marathon participations per runner.

3.1.6 Exploratory year analysis

In addition to individual runner characteristics, external factors specific to each race edition can significantly influence finishing times. Figure 8 illustrates the mean official finishing time for each year from 2014 to 2019, accompanied by a 95% confidence interval (CI) ribbon.

This reveals substantial inter-annual variability as there is a clear lack of a linear trend, with mean times fluctuating between a low of approximately 221 minutes in 2015 and a peak of nearly 234.5 minutes in 2018.

These observations underscore the importance of accounting for the year factor in the modeling process. Since individual performances are observed within specific race editions, a hierarchical year effect may capture variation associated with race-specific conditions that are not explained by individual runner characteristics.

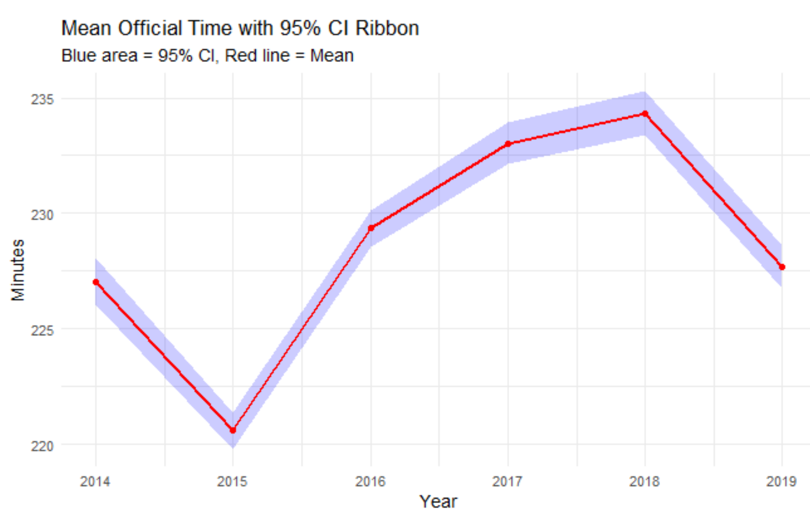

Figure 8: Distribution of official finishing times by marathon year

3.1.7 Exploratory Weather Analysis

To further investigate the year differences found in Section 3.1.6, weather information was collected from Weather Underground [2]. Four meteorological variables were considered: temperature, dew point, precipitation, and wind.

Because the Boston Marathon takes place between approximately 9:00 AM and 4:00 PM, hourly weather observations were collected during this period for both East Boston and Worcester.

For each year, average values were computed for temperature, dew point, precipitation and wind conditions. Wind measurements were transformed into a Tailwind Component using the scalar projection of the wind vector onto the overall direction of the marathon course. Positive values indicate favorable tailwind conditions, while negative values indicate headwind conditions.

Table 3 summarizes the transformed weather variables used in the exploratory analysis.

Table 3: Weather summary variables by marathon year

Year	Temperature	Dew Point	Precipitation	Tailwind Component
2014	60.75	34.56	0.0125	13.11
2015	42.63	39.94	0.0250	-13.54
2016	62.88	31.81	0.0000	-8.50
2017	67.64	37.29	0.0000	15.08
2018	39.13	38.50	0.1375	-17.25
2019	57.61	51.08	0.0213	4.68

3.1.8 Summary of the Exploratory Analysis

The exploratory analysis identified several important characteristics of the data:

- Finishing times exhibit a right-skewed distribution, motivating a logarithmic transformation.
- Age shows a nonlinear relationship with marathon performance.
- Significant differences exist between male and female runners.
- Performance varies across countries and race editions.
- A large number of runners participated in multiple years, creating a longitudinal structure.

These findings provide empirical support for the hierarchical Bayesian models introduced next.

4 Methodology

4.1 Bayesian Framework

The Bayesian approach allows us to treat model parameters as random variables, providing a full posterior distribution rather than just point estimates. This is particularly valuable in sports science, where variability between individuals and race conditions is high. Bayesian inference provides a flexible framework for uncertainty quantification and hierarchical modeling [3].

The exploratory analysis presented in Section 3.1.1 showed a right-skewed distribution of finishing times. So we applied a logarithmic transformation to obtain a more symmetric response variable and stabilize the variance.

A Generalized Additive Model for Location, Scale and Shape (GAMLSS) [4] was used to compare Normal and Student-t distributions. Based on the results, the Student-t distribution provided a substantially better fit and was therefore adopted as the likelihood for all Bayesian models.

$$\log(\text{time}_i) \sim \text{Student-t}(\nu, \mu_i, \sigma) \tag{1}$$

The log-transformation is applied to ensure the positivity of finishing times and to stabilize the variance across the wide range of performance levels.

To compare the suitability of different distributions, a Generalized Additive Model for Location, Scale and Shape (GAMLSS) was used to compare a Normal distribution against a Student-t distribution.

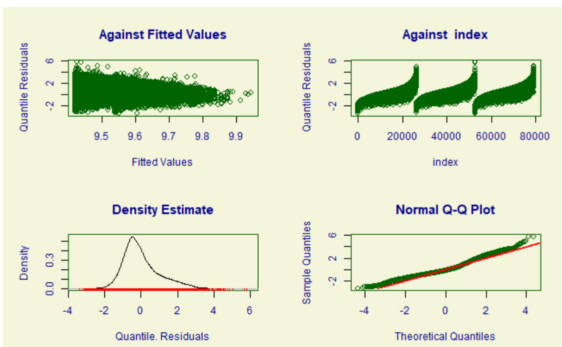

Figure 9: Diagnostic plots of Normal distribution GAMLSS model

```
AIC Normal: -71287.38
AIC Student-t: -73427.45
Estimated Degrees of Freedom (nu): 5.317079
******************************************************************
              Summary of the Quantile Residuals
                           mean   =  3.613631e-15
                       variance   =  1.000013
               coef. of skewness  =  0.8571258
               coef. of kurtosis  =  3.696485
Filliben correlation coefficient  =  0.9749375
******************************************************************
```

Figure 10: GAMLSS' results for Normal and Student-t distributions

Based on the Akaike Information Criterion (AIC), the Student-t distribution was selected as the superior model (AIC: -73,427.45) compared to the Normal distribution (AIC: -71,287.38). The estimated degrees of freedom ($\nu \approx 5.31$) confirm the presence of heavy tails, indicating that the Student-t distribution is more robust to outliers in the marathon dataset.

4.2 Baseline Model

The baseline model serves as the foundation for the analysis, focusing on the fundamental demographic drivers of performance: age and sex. Performance in endurance sports is non-linear with respect to age as seen in section 3.1.2. Athletes typically reach a peak before experiencing a gradual decline. To capture this relationship, a quadratic term for age is included.

$$\log(time_i) \sim Student\text{-}t(\nu, \mu_i, \sigma). \tag{2}$$

with

$$\mu_i = \beta_0 + \beta_{sex} * Sex_i + \beta_{age} * Age_i + \beta_{age2} * Age_i^2$$

where:

- $time_i$ is the finishing time in hours of runner i.
- Sex_i is a binary indicator equal to 1 for male runners and 0 for female runners.
- Age_i is the standardized age of the runner.

The inclusion of the quadratic age term allows the relationship between age and performance to be nonlinear: Age_i^2. This reflects that marathon performance improves until a certain age and gradually declines thereafter.

Weakly informative priors are assigned to the global parameters to allow the data to dominate the posterior while ensuring numerical stability during the sampling process.

4.3 Extended Models

The baseline model ignores potential differences between countries, race editions and individual runners. However, marathon performance may vary systematically across these groups due to cultural, environmental, and organizational factors.

To account for this structure, progressively more flexible models are introduced.

4.3.1 Model 1: Hierarchical Year Effect

As seen in section 3.1.6, Marathon performance can vary from year to year due to factors such as weather conditions, race organization or other unobserved circumstances. To account for differences between race editions, we introduce a hierarchical year effect.

Therefore, the new model is defined as

$$\log(time_i) \sim Student\text{-}t(\nu, \mu_i, \sigma)$$

with

$$\mu_i = \beta_0 + \beta_{sex}\, Sex_i + \beta_{age}\, Age_i + \beta_{age2}\, Age_i^2 + \alpha_{year[i]}.$$

The year-specific effects are modeled hierarchically as

$$\alpha_{year_j} \sim \mathcal{N}(0, \sigma_{year}), \qquad j = 1, \dots, J,$$

where J denotes the number of race editions included in the analysis.

The parameter σ_{year} represents the variability between race years. If σ_{year} is close to zero, finishing times are relatively homogeneous across years. Larger values indicate substantial differences between race editions.

This hierarchical specification implements partial pooling. To reduce overfitting and improve stability of the estimates, years with fewer observations are automatically shrunk towards the overall average, while years with more observations are allowed to deviate according to the information provided by the data.

4.3.2 Model 2: Hierarchical Country Effect

As seen in section 3.1.4, participation in the Boston Marathon is highly international. Differences in training culture, qualification standards, travel requirements, and running traditions may lead to systematic performance differences across countries.

Model 2 extends Model 1 by introducing a country-level hierarchical effect in addition to the year effect.

$$log(time_i) \mid \mu_i, \sigma \sim Student\text{-}t(\nu, \mu_i, \sigma)$$

with

$$\mu_i = \beta_0 + \beta_{sex} Sex_i + \beta_{age} Age_i + \beta_{age2} Age_i^2 + \alpha_{year[i]} + \alpha_{country[i]}.$$

The year and country effects are modeled hierarchically using a non-centered parameterization:

$$\begin{aligned} \alpha_{year,j} &= \sigma_{year} z_{year,j}, \qquad z_{year,j} \sim \mathcal{N}(0,1), \\ \alpha_{country,c} &= \sigma_{country} z_{country,c}, \qquad z_{country,c} \sim \mathcal{N}(0,1). \end{aligned}$$

The variance parameters σ_{year} and $\sigma_{country}$ quantify the amount of variability attributable to race editions and countries, respectively.

This hierarchical structure implements partial pooling. Countries with many observations obtain estimates mainly driven by their own data, while countries with fewer runners are automatically shrunk toward the overall average. This reduces overfitting and improves the stability of country-specific estimates.

Similarly, year effects capture differences between race editions that are not explained by the individual-level covariates included in the model.

4.3.3 Model 3: Hierarchical Runner Effect

The dataset contains many runners who participated in multiple editions of the Boston Marathon. To exploit this longitudinal structure, a unique `ID` was assigned to each runner. Runner identities were reconstructed by matching observations with the same Name, Sex, City, and estimated Birth Year.

Model 3 extends Model 2 by introducing a runner-specific hierarchical effect. This allows the model to account for persistent differences in performance between individuals and to model the correlation between repeated observations of the same runner.

$$log(time_i) \mid \mu_i, \sigma \sim Student\text{-}t(\nu, \mu_i, \sigma)$$

with

$$\mu_i = \beta_0 + \beta_{sex} Sex_i + \beta_{age} Age_i + \beta_{age2} Age_i^2 + \alpha_{year[i]} + \alpha_{country[i]} + \alpha_{runner[i]}.$$

The year, country, and runner effects are modeled hierarchically using a non-centered parameterization:

$$\begin{aligned}
\alpha_{year,j} &= \sigma_{year} z_{year,j}, \qquad z_{year,j} \sim \mathcal{N}(0,1), \\
\alpha_{country,c} &= \sigma_{country} z_{country,c}, \qquad z_{country,c} \sim \mathcal{N}(0,1), \\
\alpha_{runner,r} &= \sigma_{runner} z_{runner,r}, \qquad z_{runner,r} \sim \mathcal{N}(0,1).
\end{aligned}$$

The variance parameters σ_{year}, $\sigma_{country}$, and σ_{runner} quantify the variability attributable to race editions, countries, and individual runners, respectively.

This specification is the most flexible model considered in the study. By incorporating runner-specific effects, the model can separate individual performance differences from age, sex, country, and year effects. Repeated observations from the same runner contribute information to the estimation of the corresponding runner effect, leading to improved predictive performance.

The hierarchical structure also provides partial pooling, preventing overfitting for runners with only a small number of observations while still allowing substantial deviations from the population average when supported by the data.

4.4 Prior Specification

Weakly informative priors were assigned to all regression coefficients and variance parameters:

$$\begin{aligned}
\beta_0 &\sim \mathcal{N}(1.3, 0.5^2), \\
\beta_{sex}, \beta_{age}, \beta_{age2} &\sim \mathcal{N}(0,1), \\
\sigma, \sigma_{year}, \sigma_{country}, \sigma_{runner} &\sim \text{Half-Normal}(0,1), \\
\nu &\sim \text{Gamma}(2, 0.1).
\end{aligned}$$

The priors on the standard deviations ensure positivity while remaining weakly informative. The Student-t likelihood provides robustness against extreme marathon times and potential outliers in the dataset.

5 Results

5.1 Model Convergence

All fitted models showed good convergence, with $\hat{R}$ values close to 1 for the main parameters and sufficiently large effective sample sizes. Detailed trace plots and posterior density plots were inspected to verify chain mixing.

5.2 Baseline Model Results

The baseline model provides a fundamental quantification of how gender and age influence marathon finishing times. The posterior estimates for the parameters, summarized in Table 4, reflect the central tendency and uncertainty of the effects after observing the data.

Table 4: Posterior Estimates for Baseline Model Parameters

Parameter	Mean	SD	5%	95%
β_0	4.96987	0.00297	4.96505	4.97477
β_{sex}	-0.13657	0.00116	-0.13466	-0.13847
β_{age}	0.00740	0.00006	0.00731	0.00750
β_{age^2}	0.01837	0.00048	0.01758	0.01917
σ	0.09380	0.00072	0.09261	0.09501
ν	2.76867	0.05156	2.68554	2.85592

The negative coefficient for sex suggests systematic differences in performance between male and female runners. Specifically, holding age constant, male runners' finishing times are approximately 12.8% faster than those of female runners ($1 - e^{-}0.1366 \approx 0.128$). Meanwhile, the positive linear and quadratic age effects support a nonlinear relationship between age and marathon performance.This reflects the reality that the decline in performance becomes more pronounced in later decades of life. Finally, the low value of ν is highly significant as it confirms the presence of "heavy tails" in the data and the decision to use Student-t distribution.

The analysis of the MCMC diagnostics indicates excellent model performance. Figure 12's trace plots show the correctly convergence of all 4 chains. This is further supported by the Gelman-Rubin diagnostic ($\hat{R} \approx 1.00$) [5] mentioned previously. Figure 11's posterior density plots are unimodal and relatively narrow, indicating a high degree of precision in the parameter estimates due to the large sample size.

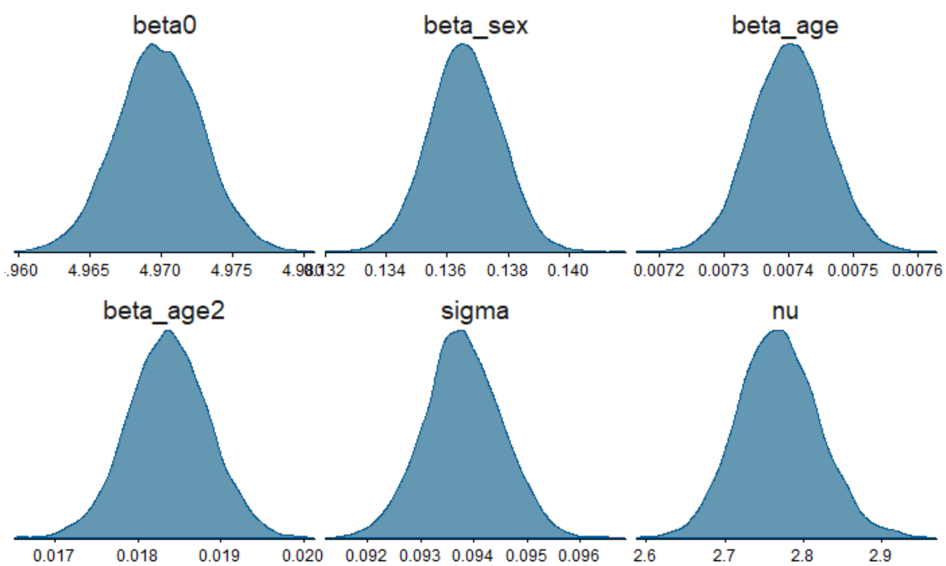

Figure 11: Posterior distributions for selected Baseline Model parameters

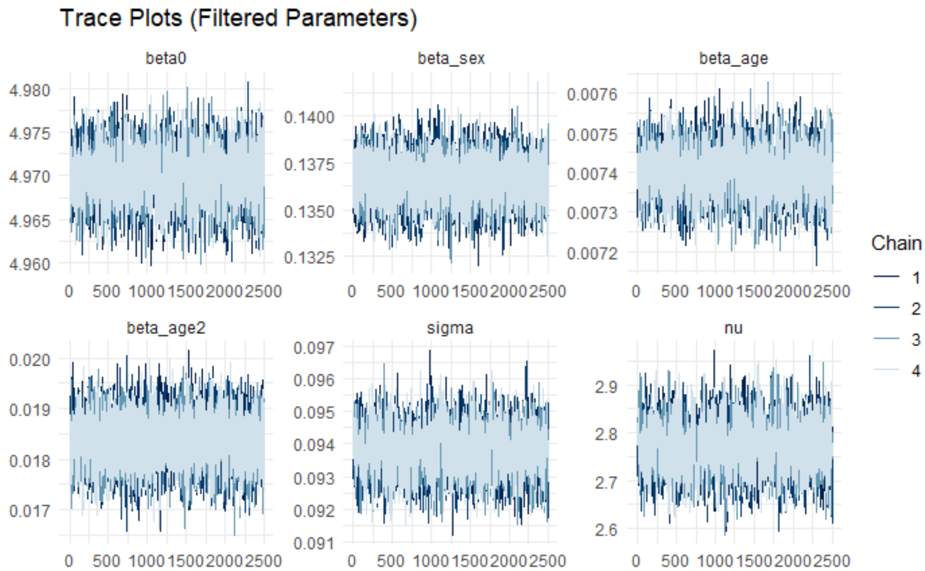

Figure 12: Trace plots for selected Baseline Model parameters

The Baseline Model confirms that while age and gender are powerful predictors of performance, there remains significant unexplained variation (σ), suggesting that the inclusion of hierarchical effects (Year, Country, and Runner ID) in subsequent models is justified to capture group-level and individual-level nuances.

5.3 Model 1: Year Effects

The estimated between-year standard deviation was $\sigma_{year} = 0.024$, indicating a modest amount of variability between race editions.

The estimated year effects suggest that some marathon editions were systematically faster or slower than average after controlling for age and sex. In particular, 2015 was associated with shorter finishing times, whereas 2017 was associated with longer finishing times.

5.3.1 Posterior Inference and Convergence Diagnostics

Table 5 summarizes the posterior estimates of the main parameters for Model 1.

Table 5: Posterior summary for Model 1

Parameter	Mean	SD	5% Quantile	95% Quantile
β_0	1.350	0.010	1.330	1.360
β_{sex}	-0.136	0.001	-0.138	-0.134
β_{age}	0.080	0.001	0.079	0.081
β_{age2}	0.018	0.0005	0.017	0.019
σ	0.092	0.001	0.090	0.093
σ_{year}	0.024	0.012	0.012	0.046

The posterior estimates for the core predictors (β_{sex}, β_{age}, and β_{age2}) remain remarkably consistent with the baseline model. This indicates that the runner's characteristics are independent of year-to-year environmental fluctuations. The estimated year-level variability was relatively small ($\sigma_{year} = 0.024$), suggesting that differences between race editions exist but explain only a limited proportion of the total variability in marathon finishing times.

Figure 13 shows the posterior distributions of the main parameters, and Figure 14 presents the corresponding trace plots.

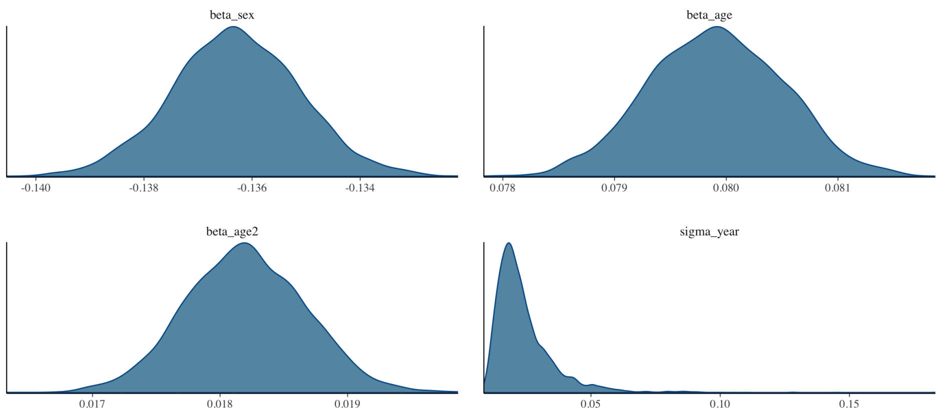

Figure 13: Posterior distributions for selected Model 1 parameters.

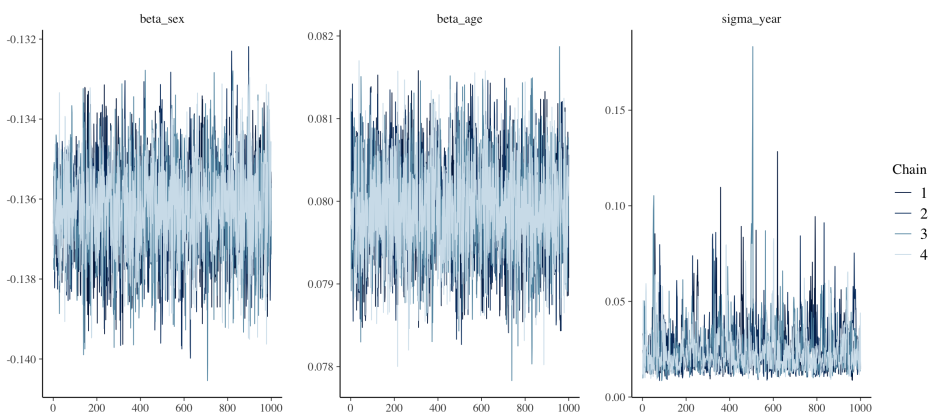

Figure 14: Trace plots for selected Model 1 parameters.

The trace plots exhibit good mixing across chains and no apparent trends, indicating satisfactory convergence of the Markov chains. This is further supported by $\hat{R}$ values close to 1 for all parameters and effective sample sizes above 500, with a good reliable posterior estimation.

Table 6: Posterior mean year effects

Year	Mean	5%	95%
2014	-0.003	-0.020	0.013
2015	-0.022	-0.039	-0.005
2016	0.013	-0.004	0.029
2017	0.023	0.006	0.040
2018	0.005	-0.012	0.022
2019	-0.016	-0.033	0.000

Among the analyzed editions, 2017 exhibited the largest positive year effect, whereas 2015 showed the most negative effect. This suggests that while the model does not explicitly include weather variables, the year-level random effects serve as a proxy for them as 2015 had had much cooler temperatures ($42.6°F$) compared to 2017, which was significantly warmer ($67.6°F$) Nevertheless, the magnitude of these differences remained modest, consistent with the relatively small estimate of σ_{year}.

Model 1 demonstrates that marathon performance is a combination of the runner's inherent characteristics and the year's transient environmental conditions. While individual age and sex remain the primary predictors, accounting for the race-day weather (captured by the year effect) is essential for a more accurate and nuanced analysis and prediction of runners' finishing times.

5.4 Model 2: Year and Country Effects

Model 2 extends the previous specification by incorporating country-level random effects. The estimated between-country variability was larger than the between-year variability. Nationality captures an important source of heterogeneity in marathon performance.

The posterior mean of the country-level standard deviation was $\sigma_{country} = 0.110$, compared to $\sigma_{year} = 0.025$. Differences between countries explain considerably more variation in finishing times than differences between race editions.

5.4.1 Posterior Inference and Convergence Diagnostics

Table 7: Posterior summary of variance parameters for Model 2

Parameter	Mean	SD	5% Quantile	95% Quantile
σ_{year}	0.025	0.016	0.011	0.055
$\sigma_{country}$	0.110	0.012	0.090	0.131

Figure 15 shows the posterior distributions of the main parameters, Figure 16 presents the corresponding trace plots.

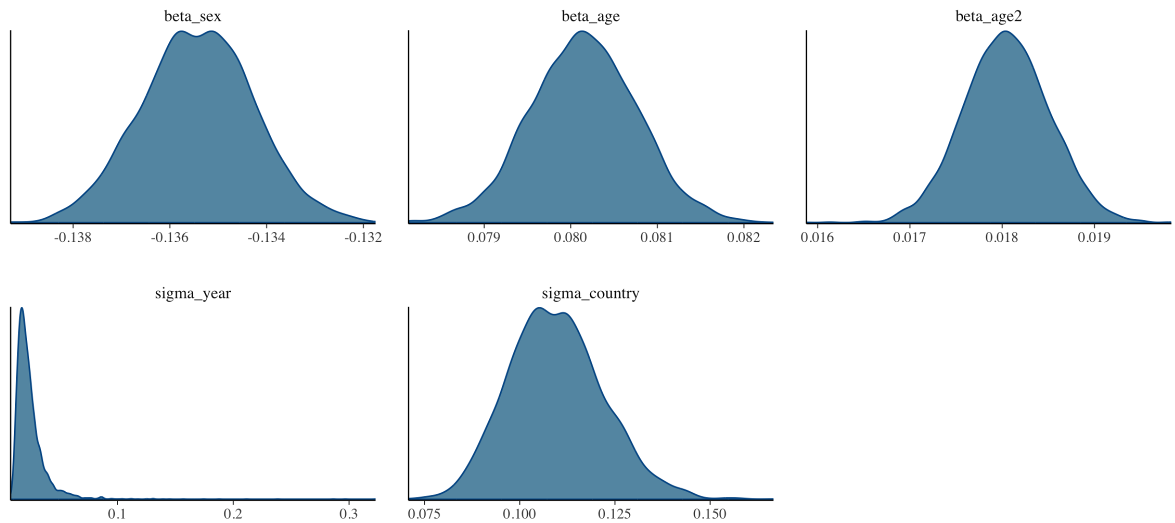

Figure 15: Posterior distributions for selected Model 2 parameters.

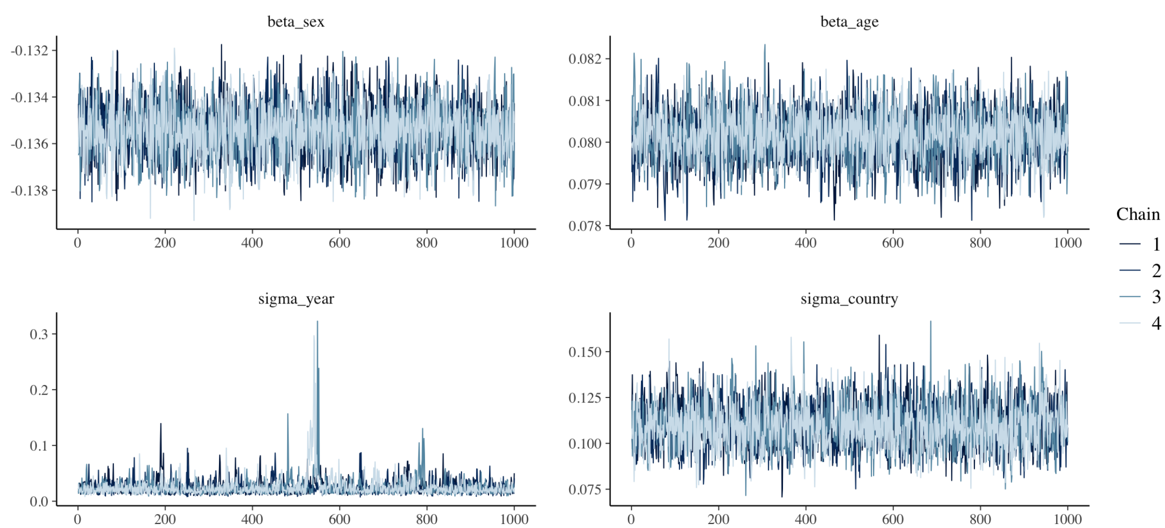

Figure 16: Trace plots for selected Model 2 parameters.

The trace plots show satisfactory mixing between chains and no visible signs of non-convergence. All monitored parameters exhibited $\hat{R}$ values close to one and sufficiently large effective sample sizes.

Table 8: Country effects estimated in Model 2

Fastest Countries

Country	Mean Effect
KEN	-0.312
ETH	-0.308
BLR	-0.210
UKR	-0.205
SLO	-0.081

Slowest Countries

Country	Mean Effect
KOR	0.451
INA	0.136
GRE	0.114
DEN	0.099
ISL	0.094

The estimated country effects reveal differences in marathon performance across nationalities. As expected, Kenya and Ethiopia exhibited the strongest negative effects, showing faster finishing times after controlling for age, sex, and race year.

Countries such as South Korea and Indonesia showed positive effects, corresponding to longer finishing times on average. These results should be interpreted cautiously, as some countries are represented by a relatively small number of runners.

Model 2 demonstrates that country-level variation represents a major source of heterogeneity in marathon performance. Compared with Model 1, the inclusion of country effects improves the model's ability to capture systematic differences between runners.

5.5 Model 3: Year, Country and Runner Effects

Model 3 extends Model 2 by incorporating runner-level random effects. This is the most flexible model considered in the analysis, as it accounts for repeated observations of the same runners across different Boston Marathon editions.

The main improvement of Model 3 is the substantial reduction in residual variability. The residual standard deviation decreased to $\sigma = 0.040$, compared with approximately $\sigma = 0.092$ in Models 1 and 2. Runner-specific effects explain an important proportion of the variability in marathon finishing times.

5.5.1 Posterior Inference and Convergence Diagnostics

Table 9: Posterior summary of main parameters for Model 3

Parameter	Mean	SD	5% Quantile	95% Quantile
β_0	1.320	0.019	1.290	1.350
β_{sex}	-0.127	0.002	-0.130	-0.123
β_{age}	0.068	0.001	0.066	0.070
β_{age2}	0.023	0.001	0.022	0.024
σ	0.040	0.0004	0.039	0.041
σ_{year}	0.023	0.012	0.012	0.045
$\sigma_{country}$	0.094	0.013	0.074	0.117
σ_{runner}	0.129	0.001	0.128	0.131

The largest variability component was the runner-level standard deviation, $\sigma_{runner} = 0.129$, followed by the country-level variability, $\sigma_{country} = 0.094$, and the year-level variability, $\sigma_{year} = 0.023$. Persistent individual differences between runners represent the main source of heterogeneity in the dataset.

Figure 17 shows the posterior distributions of the main parameters, and Figure 18 presents the corresponding trace plots.

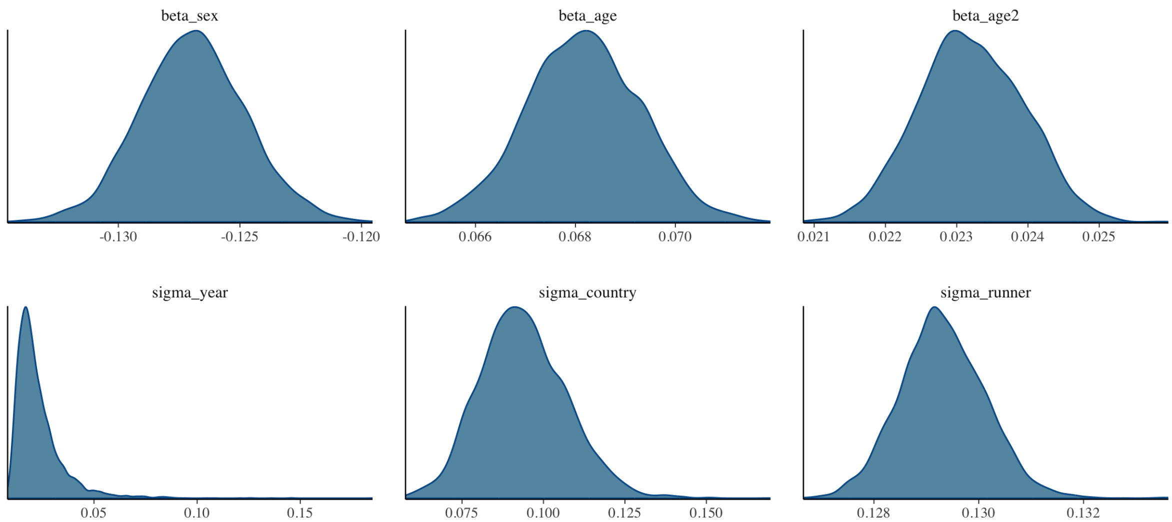

Figure 17: Posterior distributions for selected Model 3 parameters.

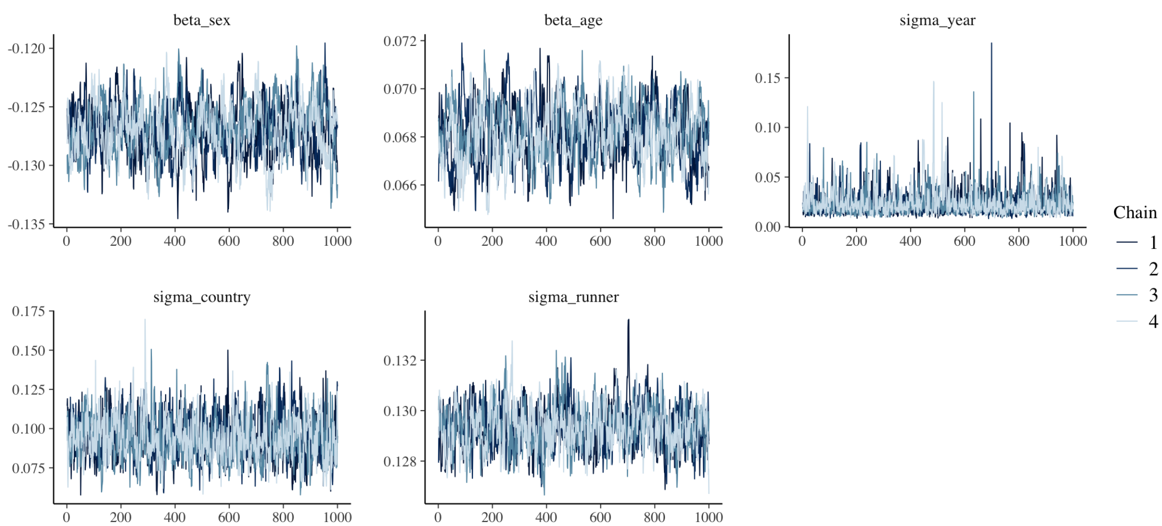

Figure 18: Trace plots for selected Model 3 parameters.

The trace plots show satisfactory mixing across chains. All monitored parameters presented $\hat{R}$ values close to one and effective sample sizes above 290.

5.5.2 Year and Country Effects

The estimated year effects were similar to those obtained in the previous models. The year 2015 was associated with shorter-than-average finishing times, and 2017 showed longer-than-average finishing times.

Table 10: Posterior mean year effects for Model 3

Year	Mean	5%	95%
2014	-0.008	-0.025	0.009
2015	-0.027	-0.045	-0.011
2016	0.007	-0.011	0.023
2017	0.018	0.001	0.034
2018	0.012	-0.005	0.028
2019	-0.005	-0.022	0.011

Country effects were also reduced compared with Model 2 after accounting for runner-specific variability. Kenya and Ethiopia remained the countries with the most negative effects, indicating faster-than-average finishing times after controlling for age, sex, year and runner-level heterogeneity.

Table 11: Country effects estimated in Model 3

Fastest Countries

Country	Mean Effect
KEN	-0.291
ETH	-0.254
BLR	-0.122
UKR	-0.113
DOM	-0.060

Slowest Countries

Country	Mean Effect
KOR	0.316
DEN	0.077
POL	0.070
MAS	0.068
INA	0.063

Model 3 provides the most complete representation of the data. By including runner-specific random effects, it captures persistent individual differences and substantially reduces unexplained variability. For this reason, Model 3 was selected as the final model for prediction.

5.6 Model Comparison

To compare the contribution of each hierarchical level, the posterior estimates of the variance components were examined across models.

Table 12 summarizes the estimated variability components for the three hierarchical models.

Table 12: Comparison of hierarchical variance components across models

Model	σ	σ_{year}	$\sigma_{country}$	σ_{runner}
Model 1	0.0915	0.0241	–	–
Model 2	0.0915	0.0248	0.1100	–
Model 3	0.0400	0.0232	0.0938	0.1290

The inclusion of country-level effects in Model 2 revealed the existence of performance differences between nationalities, with an estimated standard deviation of $\sigma_{country} = 0.1100$. However, the residual variability remained practically unchanged compared to Model 1, with $\sigma = 0.0915$ in both cases.

The introduction of runner-specific effects in Model 3 improved the model. The residual standard deviation decreased from 0.0915 to 0.0400, representing a reduction of approximately 56% in unexplained variability:

$$\frac{0.0915 - 0.0400}{0.0915} \approx 0.563.$$

This result shows that repeated observations from the same runners contain valuable information that cannot be captured solely through age, sex, year, or country effects.

Comparing the hierarchical variance components, runner-level variability was the largest source of heterogeneity in the dataset, followed by country-level variability and year-level variability:

$$\sigma_{runner} > \sigma_{country} > \sigma_{year}.$$

Model 3 provides the most realistic representation of the data, as it explicitly accounts for repeated observations of the same runners.

5.7 Predictive Performance

To evaluate the predictive ability of the final model, Model 3 was used to predict finishing times for the 2019 Boston Marathon. The model was trained using previous editions and predictions were evaluated on runners appearing in the 2019 dataset.

Table 13 summarizes the predictive performance.

Table 13: Predictive performance of Model 3 on 2019 data

Metric	Value
Number of predictions	3744
Mean Absolute Error (MAE)	9.86 minutes
Root Mean Squared Error (RMSE)	16.00 minutes
Correlation	0.920
R^2 approximation	0.846

The model achieved a mean absolute error of 9.86 minutes and a correlation of 0.920 between observed and predicted finishing times. This shows us a strong predictive performance, especially considering the natural variability of marathon races.

Figure 19 compares observed and predicted finishing times. Most observations are located close to the diagonal line, indicating good agreement between the model predictions and the actual results.

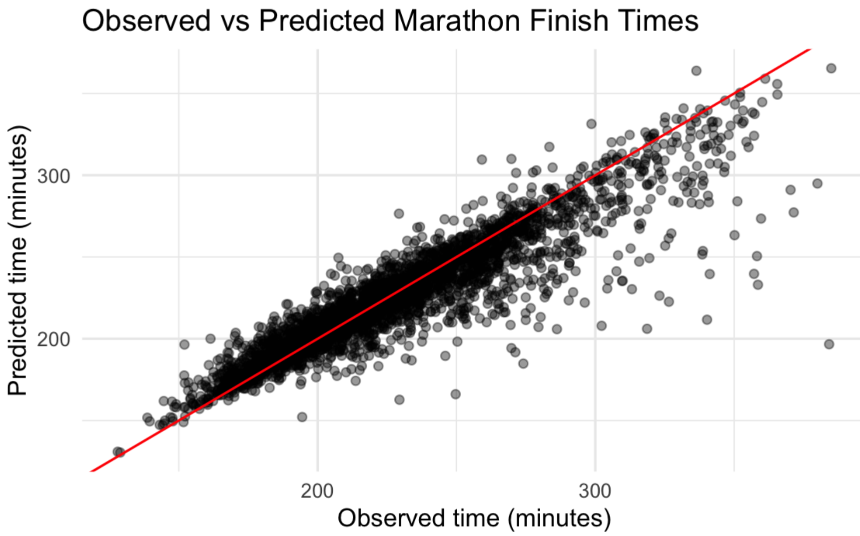

Figure 19: Observed versus predicted marathon finishing times for 2019.

Figure 20 shows the distribution of prediction errors. The distribution is centered around zero, suggesting that the model does not systematically overestimate or underestimate finishing times.

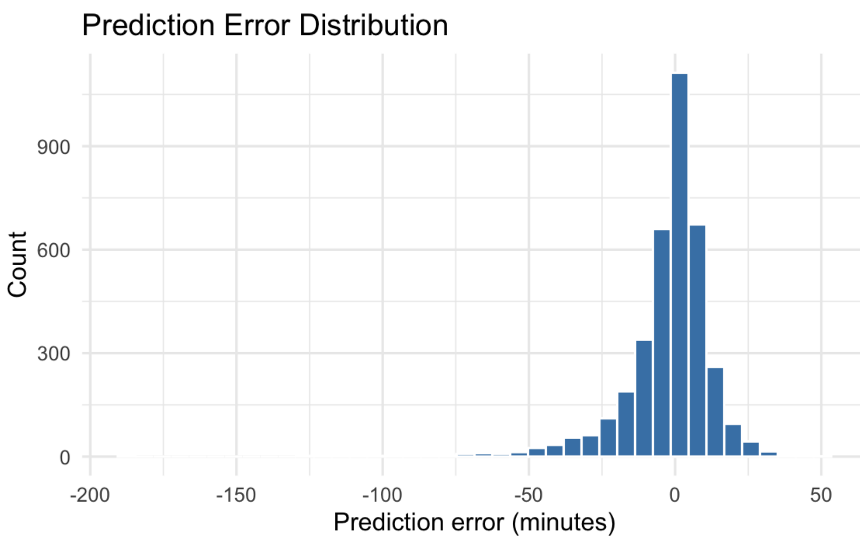

Figure 20: Distribution of prediction errors for 2019 predictions.

A further inspection of prediction errors reveals that the model performs particularly well for runners finishing below four hours, while larger errors are observed among slower runners. This pattern is visible in Figure 21, where the dispersion of prediction errors increases for longer finishing times.

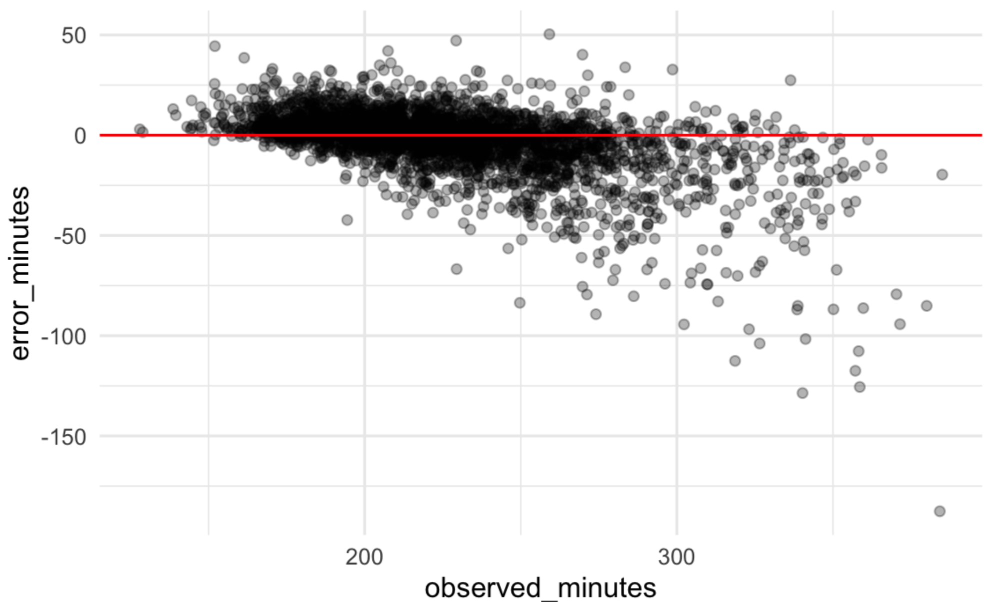

Figure 21: Prediction errors as a function of observed finishing time.

Table 14: Prediction performance by finishing-time group

Group	MAE (min)	RMSE (min)	Number of runners
Sub 3h	7.16	9.87	378
3h–4h	7.09	9.78	2266
Over 4h	16.50	25.20	1100

This result is consistent with the greater variability typically observed among recreational runners, whose race outcomes may be more affected by pacing problems, injuries, fatigue, or race-day conditions not included in the model.

6 Conclusions

This study applied Bayesian hierarchical models to analyze and predict finishing times in the Boston Marathon between 2014 and 2019. The analysis showed that marathon performance is influenced by both individual characteristics and hierarchical sources of variability, including race year, country, and repeated participation of the same runners.

The exploratory analysis confirmed that finishing times are right-skewed, motivating the use of a log-transformed response variable and a Student-t likelihood. Age and sex were found to be relevant predictors of performance, with age showing a nonlinear relationship with finishing time.

The comparison of the hierarchical models showed that year-level variability was relatively modest, while country-level effects captured more differences between runners from different nationalities. However, the largest source of heterogeneity was found at the runner level. The inclusion of runner-specific random effects reduced the residual standard deviation from approximately 0.092 to 0.040, representing a reduction of about 56% in unexplained variability.

Model 3, which included year, country, and runner-level effects, provided the most complete representation of the data and was selected as the final predictive model. When evaluated on the 2019 race results, it achieved a mean absolute error of 9.86 minutes, a root mean squared error of 16.0 minutes, and a correlation of 0.920 between observed and predicted finishing times.

The results highlight the value of Bayesian hierarchical modeling for sports performance analysis. In particular, repeated observations from the same athletes provide important longitudinal information that improves both interpretation and prediction of marathon finishing times.

7 Limitations and Extensions

While the hierarchical models developed in this study provide a robust framework for understanding marathon performance, we had some limitations regarding data composition, scope, and computational complexity. We attempted to extend the model with environmental data, with significant challenges in Bayesian estimation.

A primary limitation of the study is the potential selection bias introduced during the data cleaning phase. A significant proportion of the observations used to fit the models consists of repeated participants. This occurred because runners who participated in only a single edition often had the most incongruent or missing data fields, particularly regarding city and country of residence, making it difficult to generate reliable unique identifiers. Consequently, the results may over-represent the performance trends of consistent or more experienced marathoners, who might behave differently than one-time participants.

The dataset did not reliably distinguish between different race divisions. Specifically, wheelchair participants and handcycle athletes were not clearly flagged in all source datasets. Since these divisions follow entirely different physiological and mechanical performance curves compared to foot runners, their inclusion may introduce noise into the "heavy tails" of the distribution.

Beyond these data constraints, the scale of the hierarchical model presented a significant computational burden. With over 50,000 observations and 15,000 unique runner-level intercepts, the memory requirements and sampling time for the Markov Chain Monte Carlo (MCMC) algorithms were very big. To maintain feasibility for this assignment and ensure chain convergence within a reasonable timeframe, the total dataset size was reduced to around 40,000.

Finally, following the exploratory findings in Section 3.1.7 and the conclusions of Model 1 (4.3.1), an extension was attempted to explain the inter-annual variability (α_{year}). This was done by incorporating specific weather covariates (precipitation, temperature, and wind) as fixed effects within the hierarchical year-specific structure. The proposed extended model was specified as follows:

$$log(time_i) \mid \mu_i, \sigma \sim Student\text{-}t(\nu, \mu_i, \sigma)$$

with

$$\mu_i = \beta_0 + \beta_{sex} Sex_i + \beta_{age} Age_i + \beta_{age2} Age_i^2 + \alpha_{year[i]} + \alpha_{country[i]} + \alpha_{runner[i]}.$$

$$\begin{aligned} \alpha_{country,c} &= \sigma_{country} z_{country,c}, \qquad z_{country,c} \sim \mathcal{N}(0,1), \\ \alpha_{runner,r} &= \sigma_{runner} z_{runner,r}, \qquad z_{runner,r} \sim \mathcal{N}(0,1). \end{aligned}$$

The updated hierarchical year effect:

$$\alpha_{year,j} = \gamma_{precip}Precip_j + \gamma tempTemp_j + \gamma windWind_j + \sigma yearz_{year,j}\ z_{year,j} \qquad \sim \mathcal{N}(0,1)$$

But the attempt to fit this model encountered severe sampling issues. The model produced several divergent transitions and a maximum $\hat{R}$ of around 2, far exceeding the acceptable threshold of 1.01. The failure of this extension was likely due to the limited number of year observations ($J = 6$) as attempting to estimate three fixed weather coefficients (γ) plus a group-level variance (σ_{year}) from only six data points leads to an over-parameterized model at the group level. Consequently, this extension was omitted from the final performance analysis in favor of the more stable Model 3.

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